

2.10.85

EE25BTECH11057 - Rushil Shanmukha Srinivas

Problem: Let P be the plane $3x+2y+3z=16$ and let $S: \alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$, where $\alpha+\beta+\gamma=7$ and the distance of (α, β, γ) from the plane is $2/\sqrt{22}$. Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be three distinct vectors in S such that $|\mathbf{u}-\mathbf{v}|=|\mathbf{v}-\mathbf{w}|=|\mathbf{w}-\mathbf{u}|$. Let V be the volume of the parallelepiped determined by vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$. Then the value of $(80/3)V$ is

Solution:

$$P: \mathbf{n}^\top \mathbf{x} = c, \quad \mathbf{n} = \begin{pmatrix} 3 \\ 2 \\ 3 \end{pmatrix} c = 16. \quad (0.1)$$

The distance of point $\mathbf{P}_0 = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$ from the plane is

$$\text{dist}(\mathbf{P}_0, P) = \frac{|\mathbf{n}^\top \mathbf{P}_0 - c|}{\|\mathbf{n}\|} \quad (0.2)$$

Given $\alpha + \beta + \gamma = 7$ and $\text{dist}(\mathbf{P}_0, P) = \frac{2}{\sqrt{22}}$, we have

$$\frac{|3\alpha + 2\beta + 3\gamma - 16|}{\sqrt{22}} = \frac{2}{\sqrt{22}} \implies \mathbf{n}^\top \mathbf{P}_0 = 18 \text{ or } \mathbf{n}^\top \mathbf{P}_0 = 14. \quad (0.3)$$

Thus S lies on the intersections

$$\Pi: \mathbf{m}^\top \mathbf{x} = 7, \quad \mathbf{m} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad (0.4)$$

with

$$P_+: \mathbf{n}^\top \mathbf{x} = 18, \quad P_-: \mathbf{n}^\top \mathbf{x} = 14. \quad (0.5)$$

Each intersection is a line. The direction vector of the line is

$$\mathbf{d} = \mathbf{m} \times \mathbf{n} = \begin{pmatrix} m_{23} & n_{23} \\ m_{31} & n_{31} \\ m_{12} & n_{12} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}. \quad (0.6)$$

Pick points: For $\Pi \cap P_+$, solve

$$\begin{cases} \mathbf{m}^\top \mathbf{x} = 7, \\ \mathbf{n}^\top \mathbf{x} = 18 \end{cases} \implies \mathbf{p} = \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix} \quad (0.7)$$

So

$$L_+ : \mathbf{r} = \mathbf{p} + t\mathbf{d}. \quad (0.8)$$

For $\Pi \cap P_-$, solve

$$\begin{cases} \mathbf{m}^\top \mathbf{x} = 7, \\ \mathbf{n}^\top \mathbf{x} = 14 \end{cases} \implies \mathbf{q} = \begin{pmatrix} 0 \\ 7 \\ 0 \end{pmatrix} \quad (0.9)$$

So

$$L_- : \mathbf{r} = \mathbf{q} + t\mathbf{d}. \quad (0.10)$$

Construct equilateral triangle:

Take

$$\mathbf{u} = \mathbf{p} + s\mathbf{d}, \quad \mathbf{v} = \mathbf{p} - s\mathbf{d}, \quad \mathbf{w} = \mathbf{q} + t\mathbf{d}. \quad (0.11)$$

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u} - \mathbf{w}\|^2 = \|\mathbf{v} - \mathbf{w}\|^2. \quad (0.12)$$

$$\mathbf{u} - \mathbf{v} = 2s\mathbf{d} \implies \|\mathbf{u} - \mathbf{v}\|^2 = 4s^2\|\mathbf{d}\|^2. \quad (0.13)$$

$$\|\mathbf{d}\|^2 = \mathbf{d}^\top \mathbf{d} = 1^2 + 0^2 + (-1)^2 = 2, \quad \|\mathbf{u} - \mathbf{v}\|^2 = 8s^2. \quad (0.14)$$

$$\mathbf{r} = \mathbf{p} - \mathbf{q} = \begin{pmatrix} 0 \\ -4 \\ 4 \end{pmatrix}, \quad \mathbf{u} - \mathbf{w} = \mathbf{r} + (s - t)\mathbf{d}. \quad (0.15)$$

$$\|\mathbf{u} - \mathbf{w}\|^2 = \|\mathbf{r}\|^2 + 2(s - t)(\mathbf{r}^\top \mathbf{d}) + (s - t)^2\|\mathbf{d}\|^2. \quad (0.16)$$

$$\|\mathbf{r}\|^2 = 32, \quad \mathbf{r}^\top \mathbf{d} = -4, \quad \|\mathbf{d}\|^2 = 2. \quad (0.17)$$

$$\|\mathbf{u} - \mathbf{w}\|^2 = 32 - 8(s - t) + 2(s - t)^2. \quad (0.18)$$

$$\mathbf{v} - \mathbf{w} = \mathbf{r} + (-s - t)\mathbf{d} \implies \|\mathbf{v} - \mathbf{w}\|^2 = 32 + 8(s + t) + 2(s + t)^2. \quad (0.19)$$

Equating

$$32 - 8(s - t) + 2(s - t)^2 = 8s^2, \quad 32 + 8(s + t) + 2(s + t)^2 = 8s^2. \quad (0.20)$$

Subtracting,

$$16s + 8st = 0 \Rightarrow t = -2, s^2 = 4. \quad (0.21)$$

Take $s = 2, t = -2$:

$$\mathbf{u} = \mathbf{p} + 2\mathbf{d} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}, \quad \mathbf{v} = \mathbf{p} - 2\mathbf{d} = \begin{pmatrix} -2 \\ 3 \\ 6 \end{pmatrix}, \quad \mathbf{w} = \mathbf{q} - 2\mathbf{d} = \begin{pmatrix} -2 \\ 7 \\ 2 \end{pmatrix}. \quad (0.22)$$

Volume of parallelepiped:

Gram matrix G (matrix of pairwise inner products):

$$G = \begin{pmatrix} \|\mathbf{u}\|^2 & \mathbf{u}^\top \mathbf{v} & \mathbf{u}^\top \mathbf{w} \\ \mathbf{v}^\top \mathbf{u} & \|\mathbf{v}\|^2 & \mathbf{v}^\top \mathbf{w} \\ \mathbf{w}^\top \mathbf{u} & \mathbf{w}^\top \mathbf{v} & \|\mathbf{w}\|^2 \end{pmatrix} = \begin{pmatrix} 17 & 17 & 21 \\ 17 & 49 & 37 \\ 21 & 37 & 57 \end{pmatrix}, \quad (0.23)$$

Determinant of the Gram matrix (gives the squared volume):

$$\det G = \det \begin{pmatrix} 17 & 17 & 21 \\ 17 & 49 & 37 \\ 21 & 37 & 57 \end{pmatrix} = 12544 \quad (0.24)$$

Hence the volume itself is

$$V = \sqrt{\det G} = \sqrt{12544} = 112. \quad (0.25)$$

now

$$\frac{80V}{3} = \frac{80 * 112}{3} = \frac{8960}{3}. \quad (0.26)$$

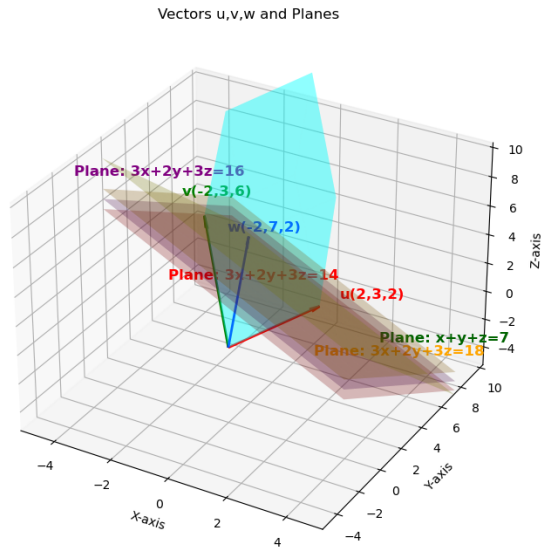


Fig: Representation of Planes and vectors